



Catholic Junior College

JC2 Preliminary Examinations

Higher 1

CANDIDATE
NAME

CLASS

2T

CHEMISTRY

Structured Questions

8873/02

17 August 2018

2 hours

Candidates answer on the Question Paper.
Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A – Answer **all** the questions.
Section B – Answer **one** question.

The use of an approved scientific calculator is expected where appropriate.
You are reminded of the need for good English and clear presentation in your answers.

A Data Booklet is provided.

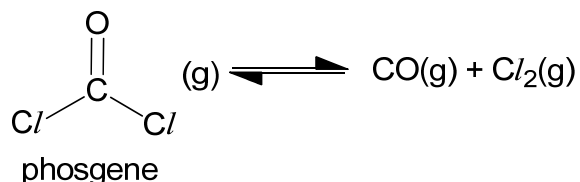
The number of marks is given in brackets [] at the end of each question or part of the question.

Section A	Q1	12
	Q2	6
	Q3	15
	Q4	6
	Q5	6
	Q6	15
Section B	Q7	20
	Q8	20
Paper 2		80
Paper 1		30
Percentage		
Grade		

Section A

Answer **all** the questions in this section in the spaces provided.

- 1 (a)** Phosgene, COCl_2 is a gas with many industrial uses. At high temperature, gaseous phosgene decomposes to carbon dioxide and chlorine in a dynamic equilibrium according to the following equation:



- (i)** Explain what is meant by the term *dynamic equilibrium*.

.....

[1]

- (ii)** Write an expression for the equilibrium constant, K_c , for the reaction.
 Calculate its value when, at equilibrium, the concentrations of $\text{COCl}_2(\text{g})$, $\text{CO}(\text{g})$, and $\text{Cl}_2(\text{g})$ are $0.219 \text{ mol dm}^{-3}$, $0.530 \text{ mol dm}^{-3}$ and $0.530 \text{ mol dm}^{-3}$ respectively.

[2]

- (iii)** A certain amount of phosgene was added and the new equilibrium concentration of phosgene is $0.290 \text{ mol dm}^{-3}$. Calculate the new equilibrium concentration of $\text{CO}(\text{g})$.

[1]

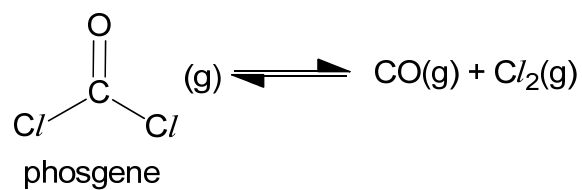
- (iv) Calculate the amount in moles of phosgene added in part (iii), given that the volume of the container is 3 dm³.

[2]

- (b) Predict and explain how a decrease in pressure of the system will affect the position of equilibrium.

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.....[2]

- (c) (i) By using bond energy values from the *Data Booklet*, calculate the enthalpy change for the decomposition of phosgene as shown below:



[2]

- (ii) Use your answer from part (i) to sketch a reaction pathway diagram for the decomposition of phosgene. Draw a labelled arrow to show the enthalpy change on your diagram.

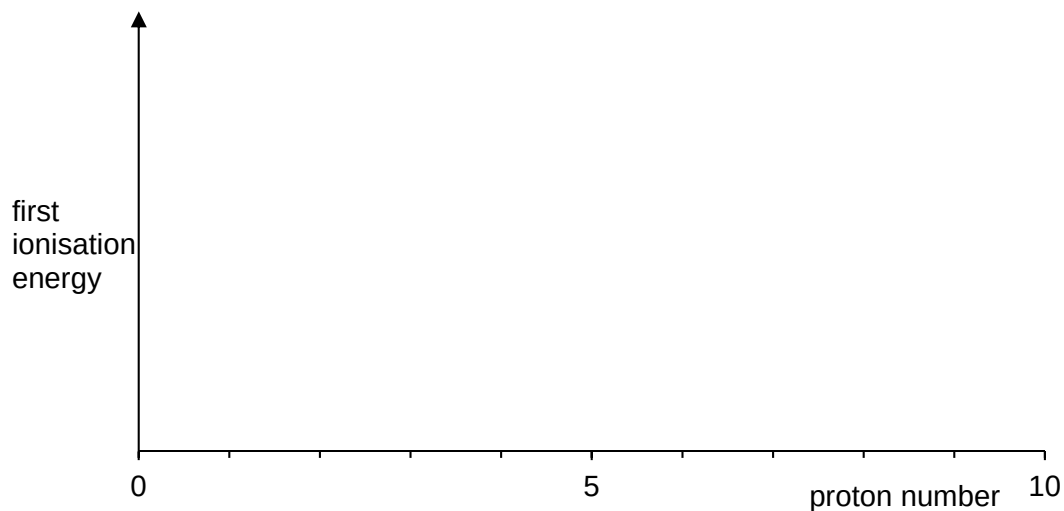
[2]

[Total: 12]

2(a) Write an equation to show the first ionisation energy of oxygen.

.....[1]

(b) Sketch how the first ionisation energies of the elements change from lithium to fluorine.



[1]

(c) Explain why the first ionisation energy of oxygen is lower than

- nitrogen

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.....[2]

- fluorine

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.....[2]

[Total: 6]

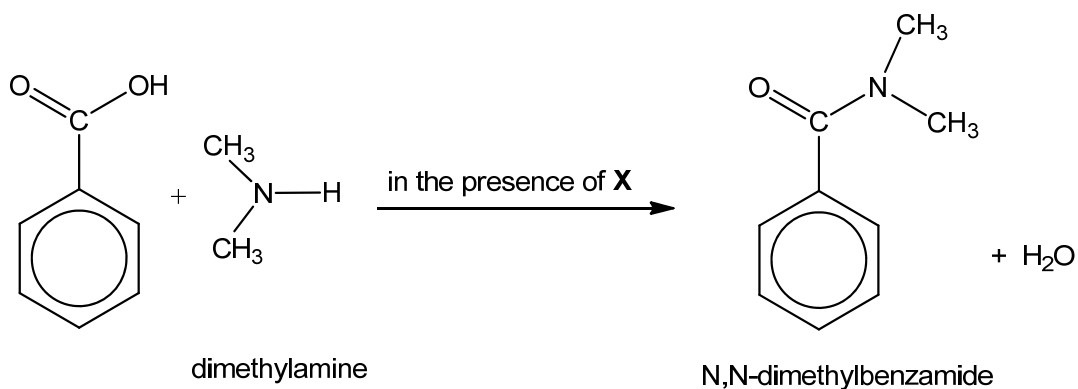
O=C(O)c1ccccc1.[Na]>>[Na]OC(=O)c1ccccc1

benzoic acid + Na → sodium benzoate + 1/2 H₂

[2]

.....[4

- (b) Benzoic acid can also react with amines to produce amides as shown in the following equation.



- (i) Name the type of reaction shown by the above equation.

.....[1]

- (ii) Identify **X** which is required for the above reaction to occur.

.....[1]

- (iii) State the shape and bond angle around the N atom in dimethylamine.

Shape:

Bond angle:

[1]

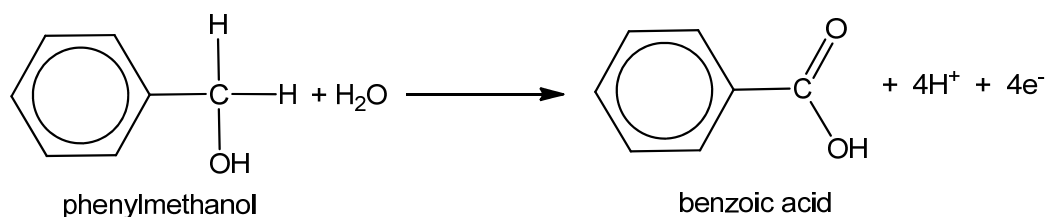
- (iv) Suggest a reason why the reaction above would **not** occur if dimethylamine is replaced by trimethylamine, $(\text{CH}_3)_3\text{N}$.

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.....[1]

- (c) Benzoic acid can be produced by oxidising phenylmethanol, $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$. A suitable oxidising agent for this reaction is hot acidified KMnO_4 .

The half equation for the oxidation of phenylmethanol is as follows:



- (i) State the colour change observed when phenylmethanol reacts with hot acidified KMnO_4 .

.....[1]

- (ii) Use the *Data Booklet* and the half equation given above to construct a balanced equation for the oxidation of phenylmethanol by hot acidified KMnO_4 .

.....[1]

- (iii) A pipette was used to transfer 25.0 cm^3 of 0.15 mol dm^{-3} of phenylmethanol into a conical flask. Calculate the minimum volume of 0.20 mol dm^{-3} of acidified KMnO_4 required to completely oxidise the phenylmethanol in the conical flask.

[2]

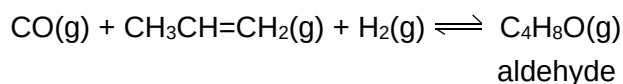
- (iv) Draw the structure of the product formed when phenylmethanol reacts with acidified $\text{K}_2\text{Cr}_2\text{O}_7$ with immediate distillation (controlled oxidation).

[1]

[Total: 15]

- 4 Carbon monoxide is a reagent used in an industrial process called hydroformylation. During hydroformylation, high pressure is used and the carbon monoxide reacts with an alkene and hydrogen gas to form an aldehyde.

The equation below shows an example of this process.



- (a) In the space below, draw the two possible constitutional isomers of $\text{C}_4\text{H}_8\text{O}$ that are produced in the reaction above.

[2]

- (b) The hydroformylation process is important because aldehydes are easily converted into many products. State the type of reaction that has occurred when an aldehyde is converted to a carboxylic acid.

.....[1]

- (c) While the hydroformylation process uses rhodium catalyst in its soluble form, elemental rhodium is used in its nanoparticle form as a *heterogeneous catalyst* for other reactants. Explain the term *heterogeneous catalyst* as fully as you can. Briefly explain why rhodium in its nanoparticle form works as a better catalyst than in its bulk form.

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.....[3]

[Total: 6]

- 5** Poly(diallyl phthalate) is an example of a cross linked, thermoset polymer. It is suitable for use in high-performance military electrical components because of its ability to retain its properties when subjected to extreme environmental conditions of high temperature and high humidity.

- (a) (i)** Draw the structure of the monomer of poly(diallyl phthalate) and state the functional group that allows the monomer to become a cross-linked polymer.

[2]

- (ii)** Explain, with reference to its structure and bonding, why it is able to retain its properties under the extreme environmental conditions stated above.

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.....[2]

- (b)** Calcium alginate is also a cross-linked polymer but its properties vary greatly from poly(diallyl phthalate). Each calcium ion can attach to two alginate polymer strands to form cross-links. Calcium alginate is used in wound dressings. When the dressing needs to be changed, the wound can be rinsed with a solution that will break the cross-links. Suggest a suitable solution that can be used and briefly explain why the cross links are broken.

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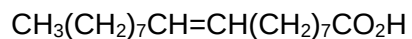
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[Total: 6]

- 6 Oleic acid is a type of fatty acid that is present in avocados. Oleic acid has the following structure:



Oleic acid has the cis configuration around the C=C double bond, whereas elaidic acid is the trans isomer of oleic acid.

- (a) Draw the condensed structural formulae of oleic acid and elaidic acid, showing clearly the configuration about the C=C in each acid.

oleic acid:

elaidic acid:

[2]

Avocado flesh turns brown rapidly upon exposure to oxygen in the air. In the presence of oxygen, enzymes in the avocado help in the formation of a class of compounds called quinones. Quinones are capable of producing polymers called polyphenols that give the brown colour.

- (b)(i)** Suggest a simple method that can be done at home to slow down the browning of avocado flesh. Give a brief explanation to support your suggestion.

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[1]

- (ii)** Define the term, *polymer*.

.....
[1]

- (iii)** 1,2-benzoquinone is an example of a quinone. It contains 66.7% carbon and 3.7% hydrogen. The rest of the molecule comprises of oxygen which is present as two ketone functional groups in a molecule of 1,2-benzoquinone.

Calculate the empirical formula of 1,2-benzoquinone. Hence, deduce the molecular formula of 1,2-benzoquinone, stating clearly how you arrived at your answer.

[3]

- (iv)** In a molecule of 1,2-benzoquinone, the two ketone functional groups are on adjacent carbon atoms in a cyclic structure. Using this information and your answer in **(iii)**, draw the skeletal structure of 1,2-benzoquinone.

[1]

Avocados ripen faster if they are put in a plastic bag with some bananas. Bananas produce trace amounts of ethene gas which functions as a ripening hormone in fruits.

- (c) Draw a dot-and-cross diagram to show the bonding in ethene. Describe this bonding in terms of orbital overlap.

[4]

Bananas are usually picked from their plantations when they are unripe. But the production of ethene can quickly ripen the fruits during transportation. Hence, to slow down the ripening process, 1-methylcyclopropene is used because it binds tightly to the ethene receptor in the fruits and therefore blocks the effect of ethene.

- (d) (i) Draw the structure of 1-methylcyclopropene.

[1]

- (ii) Explain why 1-methylcyclopropene has a higher boiling point than ethene.

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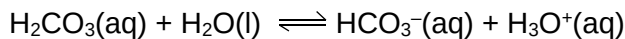
[2]

[Total: 15]

Section B

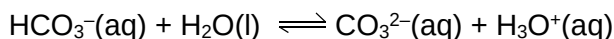
Answer **one** question in the spaces provided.

- 7(a)** Carbonic acid is a weak acid that partially dissociates in water according to the following equation:



The K_a of carbonic acid is $4.5 \times 10^{-7} \text{ mol dm}^{-3}$.

The hydrogencarbonate ion, $\text{HCO}_3^-(\text{aq})$, can also behave as weak acid in water:



The K_a of hydrogencarbonate is $5.0 \times 10^{-11} \text{ mol dm}^{-3}$.

- (i)** State the relationship between carbonic acid and the hydrogencarbonate ion.

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 [1]

- (ii)** Write the K_a expression for hydrogencarbonate ion and briefly explain why this K_a value is much smaller than the K_a value for carbonic acid.

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 [2]

- (iii)** The hydrogencarbonate ion is part of the buffer in the blood. Define *buffer* and explain how $\text{HCO}_3^-(\text{aq})$ reacts with acid in the blood. Support your answer with a suitable equation.

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 [2]

- (b) The hydroxyl radical, represented by $\bullet\text{OH}$, is highly reactive and short lived. It is often referred to as the "detergent" of the atmosphere because it reacts with greenhouse gases such as methane.



A study of the rate of this reaction in a mixture, gave the following results. The initial concentration of methane for this study was much larger than concentration of hydroxyl radical and assumed to be constant throughout the reaction.

Time after start of reaction / $\times 10^{-4} \text{ s}$	relative $[\bullet\text{OH}]$
0.0	10.0
1.0	7.0
2.0	5.0
3.0	3.5
4.0	2.5
5.0	1.8

- (i) Define the term *half-life* of a reaction.
Use the data in the table to deduce the half-life of the reaction and to show that the reaction is first order with respect to $[\bullet\text{OH}]$.

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[3]

- (ii) Assume that the order of reaction with respect to $[\text{CH}_4]$ is 1, state how the half-life of the reaction would change if:

Initial $[\text{CH}_4]$ is doubled:

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Initial relative $[\bullet\text{OH}]$ is halved:

.....[2]

- (c) Describe reactions that illustrate the variation in acid-base behaviour of the oxides of the elements in Period 3, using MgO , Al_2O_3 , and SO_3 as examples.

Write equations for all the reactions you describe.

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.....[5]

- (d) Alkane **A**, C_4H_{10} , reacts with bromine under UV light to produce only two different mono-brominated products, **B** and **C** in the likely ratio of 9:1. When reacted with alcoholic KOH , both **B** and **C** produce hydrocarbon **D**, C_4H_8 .

(i) Suggest structures for **A**, **B**, **C** and **D**. Label the structures clearly.

[3]

(ii) Describe a simple chemical test to distinguish between **A** and **D**.

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.....[2]

[Total: 20]

8(a) The table below gives the acid dissociation constants, K_a , of three acids at 25 °C.

Acid	Formula	$K_a / \text{mol dm}^{-3}$
Methanoic acid	HCO_2H	1.8×10^{-4}
Ethanoic acid	$\text{CH}_3\text{CO}_2\text{H}$	1.5×10^{-5}
Chloroethanoic acid	$\text{CH}_2\text{ClCO}_2\text{H}$	1.3×10^{-3}

- (i) From the K_a values, identify the strongest of the three acids and write the K_a expression for the acid you identified.

[2]

- (ii) A student needed to select one of the three acids above to prepare a buffer solution of pH 3.80. The acid chosen has to have a $\text{p}K_a$ that is closest to the pH value of the buffer solution.

Define the term *buffer*.

Identify the acid chosen for the preparation of the buffer and briefly explain how the buffer can be prepared (calculation of actual amounts is not required.)

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..... [3]

- (b)** 2-bromopropane, $\text{CH}_3\text{CHBrCH}_3$, can be hydrolysed by KOH(aq) to form propan-2-ol. Results of an investigation into the kinetics of this reaction are given below.

Expt number	$[\text{CH}_3\text{CHBrCH}_3] / \text{mol dm}^{-3}$	$[\text{KOH}] / \text{mol dm}^{-3}$	Relative initial rate
1	0.10	0.20	1.00
2	0.60	0.40	6.00
3	0.30	0.20	3.00

- (i) Define the term *order of reaction*.
Use the data in the table to deduce the order of reaction with respect to each reagent.

[3]

- (ii)** Write an overall rate equation for the reaction between $\text{CH}_3\text{CHBrCH}_3$ and KOH and state the units of the rate constant.

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.....[2]

- (c) The oxides Na_2O , Al_2O_3 , and P_4O_{10} differ considerably in their chemical properties. Describe what happens when separate samples of each oxide are added to water. Give equations where appropriate and state clearly why if no reactions occur.

What is the effect of adding universal indicator to each resulting solution?

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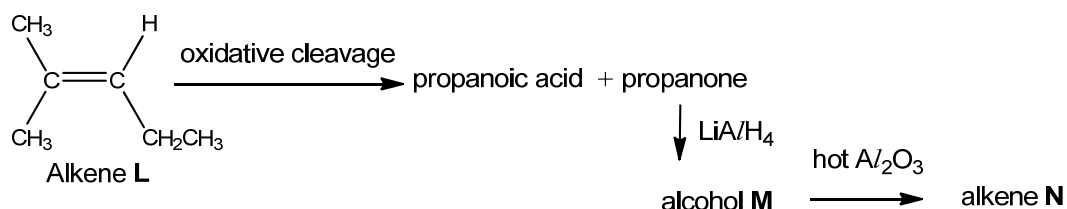
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.....[5]

- (d) When reacted with a strong oxidising agents under high temperature, alkenes can undergo *oxidative cleavage* reactions. Such reactions involve the complete breaking of the C=C bonds in the alkenes to give carbon-oxygen bonds such as C-O or C=O. These new bonds are part of functional groups such as carboxylic acids and ketones.

Alkene **L** undergoes oxidative cleavage to produce propanoic acid and propanone. Propanone is able to undergo further reactions. The reaction scheme is shown below.



- (i) Deduce the structures of **M** and **N**. [2]

Propanoic acid produced from alkene **L** reacts with alcohol **M** to give a sweet smelling organic compound.

- (ii) State the type of reaction that has taken place.
[1]

- (iii) Draw the structure of the organic product produced. [1]

- (iv) Draw the structure of the organic product produced when alkene **N** undergoes oxidative cleavage. [1]

[Total: 20]